

CHE 891 s302

Machine Learning for Chemical Process Engineering

Spring 2026

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Course Logistics

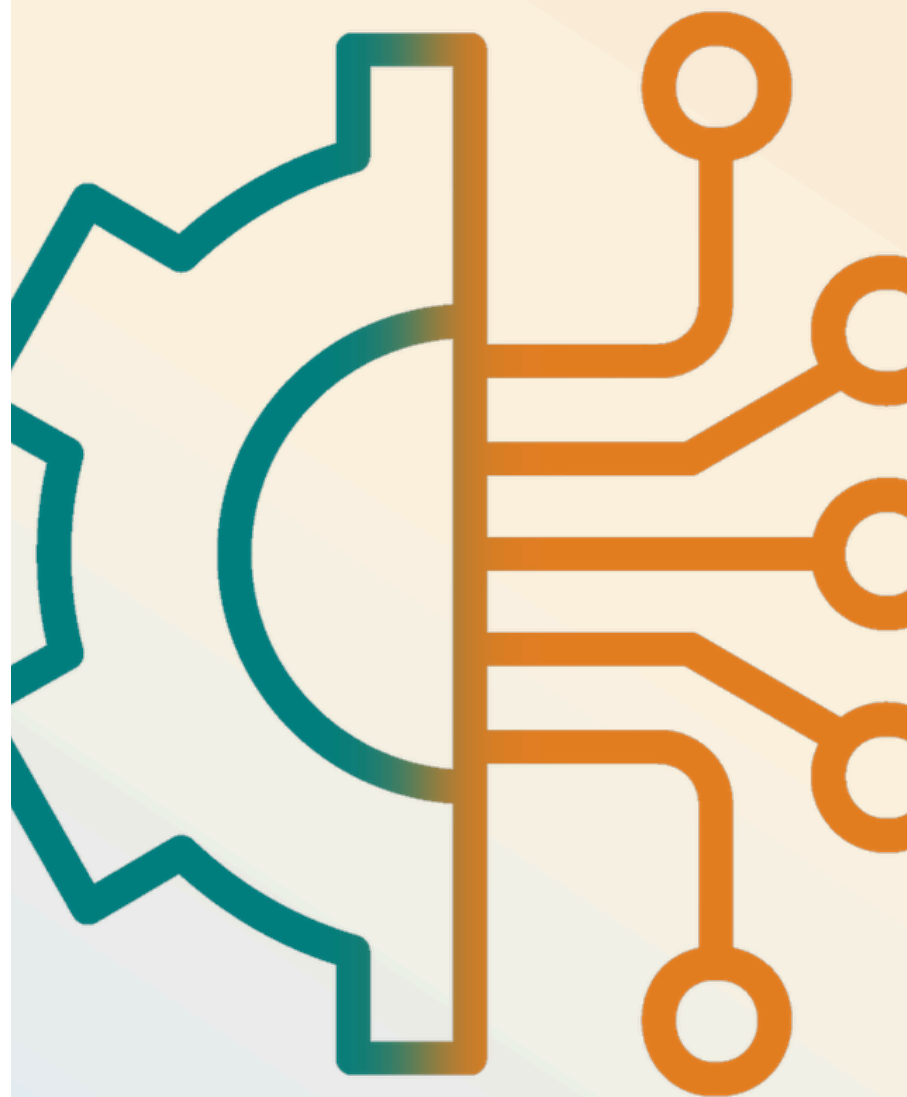
Dates	February 17 – March 20, 2026
Tuesdays	12:40 – 2:00 pm, 2245 EB
Fridays	11:00 am – 12:20 pm, 2245 EB
Office Hours	Tuesdays 2 – 3 pm, 2245 EB (or by appointment)
Contact	scb@msu.edu · 517-355-0222
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Course Description

This course applies foundational supervised and unsupervised machine learning approaches to **chemical processes**.

Process knowledge and industrial datasets are integrated with statistical learning for real-world deployment challenges.

Central theme: the spectrum of hybrid modeling approaches — from online bias correction to physics-informed neural networks — that embed process knowledge into data-driven models.



Topics

- **Unsupervised learning** and **clustering** for identifying operational regimes
- **Dimensionality reduction** for process monitoring
- **Variable selection** and visualization of high-dimensional process spaces
- Neural networks for dynamic process modeling
- Custom loss functions with **physics-based constraints**
- **Physics-informed** neural networks (PINNs)

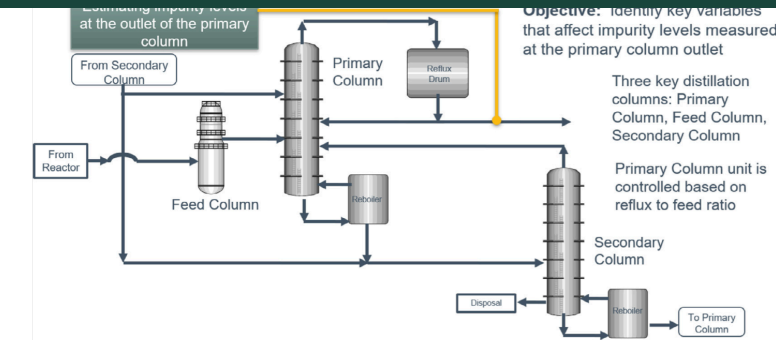


Fig. 1. Block diagram of the Dow data challenge problem considered for the industrial data analytics study.

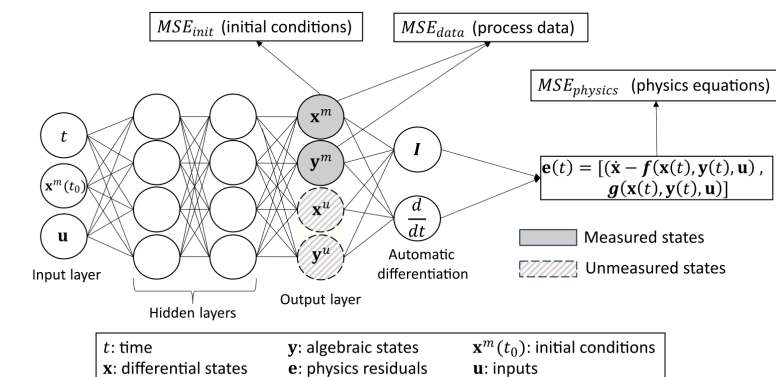
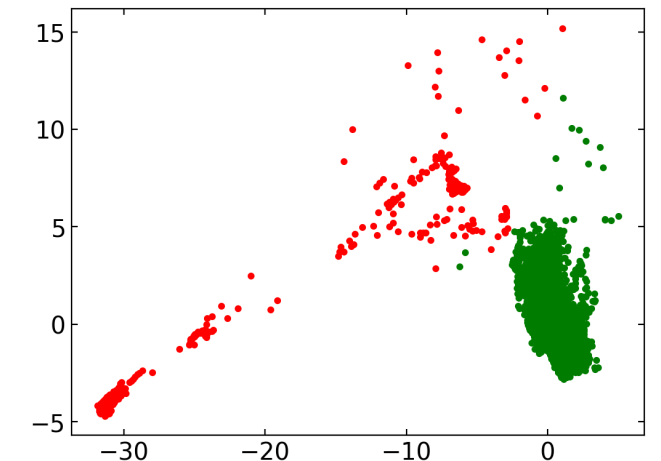


Fig. 2. PINN-based dynamic process model with semi-explicit DAE physics model.

Schedule

Week	Dates	Topics
1	Feb 17, 20	Unsupervised learning, clustering, PCA, PLS
2	Feb 24, 27	Variable selection (LARS), t-SNE/UMAP, online bias learning
—	<i>Mar 2–6</i>	<i>Spring Break</i>
3	Mar 10, 13	Neural networks, custom loss functions, hybrid models
4	Mar 17, 20	Physics-informed neural networks

Prerequisites

Machine Learning:

- Introductory ML course in Python
- Data preprocessing, supervised learning (regression/classification), model evaluation, hyperparameter tuning

Chemical Engineering:

- Process dynamics and control
- Mass and energy balances, reaction kinetics

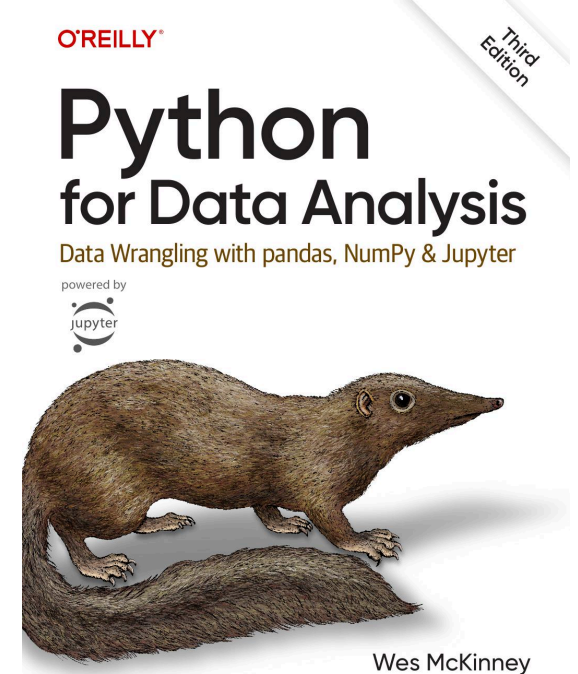
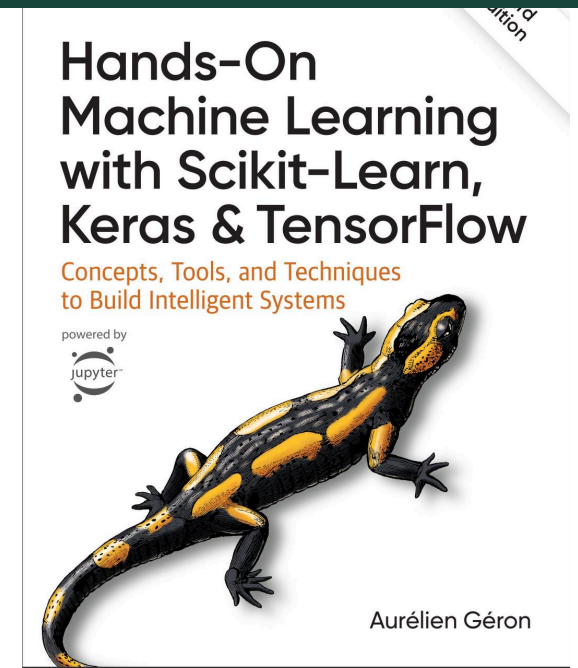
Software

Python 3.12 and Jupyterlab with:

Package	Use
<code>numpy</code> , <code>scipy</code>	Numerical computation
<code>pandas</code>	Data handling
<code>matplotlib</code>	Visualization
<code>scikit-learn</code>	Classical ML models
<code>tensorflow</code>	Neural networks, automatic differentiation

Optional Texts

- A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, 3rd Ed., O'Reilly, 2023
 - github.com/ageron/handson-ml3
- W. McKinney, *Python for Data Analysis*, 3rd Ed., O'Reilly, 2022
 - wesmckinney.com/book



Grading

Component	Weight
Homework	20%
Project	40%
Class participation	40%

Score	85–100	80–85	70–80	60–70
Grade	4.0	3.5	3.0	2.5



Project (Weeks 3–4)

Choose a process system (Van de Vusse CSTR, bioreactor, or similar)

- **Phase 1 (Week 3):** Train a neural network with a physics-constrained loss function
- **Phase 2 (Week 4):** Expand the constraint to a full governing equation (PINN)

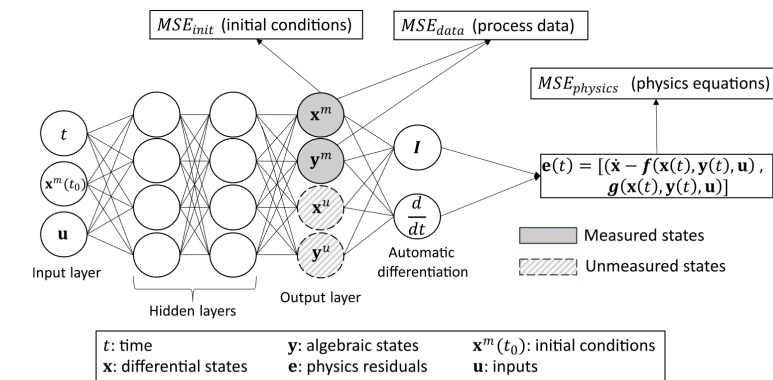


Fig. 2. PINN-based dynamic process model with semi-explicit DAE physics model.

Appropriate Use of Generative AI

Generative AI tools (ChatGPT, Claude, Copilot, etc.) are mainly helpful for:

- Troubleshooting code
- Language syntax reference
- Grammar and style checking

Strongly discouraged for drafting code that implements the ML libraries (TensorFlow, sklearn) that are the subject of this course.

Cite any AI use in your work.

The Big Picture

From Data to Physical Models

Pure data $\iff$ Hybrid $\iff$ Pure physics

	Data-driven	Hybrid	Physics-informed
Data needed	Large	Moderate	Small
Extrapolation	Poor	Good	Best
Domain knowledge included	None	Partial	Full
Interpretability	Low	Medium	High

This course walks across this spectrum, week by week.